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System and method for exhaust gas recirculation and injection in gas turbine

Abstract

A system includes a gas turbine system having a gas turbine engine including a compressor, a combustor, and a turbine. The gas turbine system further includes an exhaust gas recirculation (EGR) system configured to recirculate a first portion of an exhaust flow from the gas turbine engine to an intake of the gas turbine engine. The gas turbine system further includes an exhaust gas injection system, including a multi-stage compressor including a plurality of stages, wherein the multi-stage compressor is configured to compress a second portion of the exhaust flow. The exhaust gas injection system further includes an intercooler coupled to the multi-stage compressor. The exhaust gas injection system is configured to inject the second portion of the exhaust flow into the gas turbine engine at one or more exhaust injection locations other than the intake to provide power augmentation of the gas turbine engine.

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Primary Examiner: Sutherland; Steven M*Attorney, Agent or Firm:* Fletcher Yoder, P.C.**Background/Summary****TECHNICAL FIELD**

(1) The present application relates generally to a system and method for recirculation and injection of exhaust gas in a gas turbine system.

BACKGROUND

(2) An industrial plant, such as a combustion-driven power plant, may produce a variety of gases, such as an exhaust gas of a combustion system. The combustion system may include a gas turbine engine. These exhaust gases may include one or more undesirable gases, such as carbon containing gases. With various regulations and environmental concerns regarding global warming, it would be desirable to reduce the output of undesirable gases into the atmosphere. Unfortunately, exhaust gas treatment, such as gas capture (e.g., carbon capture), can reduce the efficiency and power output of the industrial plant. Therefore, a need exists to reduce the undesirable gases while offsetting the losses in efficiency and power output associated with the exhaust gas treatment.

BRIEF DESCRIPTION

(3) Certain embodiments commensurate in scope with the originally claimed subject matter are summarized below. These embodiments are not intended to limit the scope of the claimed embodiments, but rather these embodiments are intended only to provide a brief summary of possible forms of the subject matter. Indeed, the presently claimed embodiments may encompass a variety of forms that may be similar to or different from the embodiments set forth below.

- (4) A system includes a gas turbine system having a gas turbine engine including a compressor, a combustor, and a turbine. The gas turbine system further includes an exhaust gas recirculation (EGR) system configured to recirculate a first portion of an exhaust flow from the gas turbine engine to an intake of the gas turbine engine. The gas turbine system further includes an exhaust gas injection system, including a multi-stage compressor including a plurality of stages, wherein the multi-stage compressor is configured to compress a second portion of the exhaust flow. The exhaust gas injection system further includes an intercooler coupled to the multi-stage compressor. The exhaust gas injection system is configured to inject the second portion of the exhaust flow into the gas turbine engine at one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine.
- (5) A system includes a controller having a memory, a processor, and instructions stored on the memory and executable by the processor to control a first portion of an exhaust flow from a gas turbine engine through an exhaust gas recirculation (EGR) system to an intake of the gas turbine engine, wherein the gas turbine engine includes a compressor, a combustor, and a turbine. The controller is further configured to control a second portion of the exhaust flow from the gas turbine engine through an exhaust gas injection system to one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine.
- (6) A method including controlling a first portion of an exhaust flow from a gas turbine engine through an exhaust gas recirculation (EGR) system to an intake of the gas turbine engine, wherein the gas turbine engine includes a compressor, a combustor, and a turbine. The method further includes controlling a second portion of the exhaust flow from the gas turbine engine through an exhaust gas injection system to one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other features, aspects, and advantages of the presently disclosed techniques will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:
- (2) FIG. 1 is a block diagram of an embodiment of a system having a gas turbine system, one or more gas capture systems, and an exhaust gas recirculation (EGR) system;
- (3) FIG. 2 is a schematic of an embodiment of the system of FIG. 1, further illustrating an exhaust gas injection system configured to inject exhaust gas from the EGR system into one or more locations of the gas turbine system; and
- (4) FIG. 3 is a flowchart of an example process for operating the exhaust gas injection system of FIG. 2.

DETAILED DESCRIPTION

- (5) One or more specific embodiments of the presently disclosed systems and methods are described below. In an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development

effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure. (6) When introducing elements of various embodiments of the presently disclosed embodiments, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

(7) The disclosed embodiments include systems and methods to increase the power output and efficiency of a gas turbine system by injecting a portion of the exhaust gas of the EGR system of the gas turbine system into a compressor discharge casing of a gas turbine engine of the gas turbine system. Prior to injection into the compressor discharge casing, the portion of the exhaust gas is compressed via a compressor that is powered by a reciprocating engine. In certain embodiments, the compressor is a multi-stage compressor having a plurality of compression stages with intercoolers between the compression stages. Additionally, the reciprocating engine may provide waste heat to thermally adjust a temperature of the exhaust gas being injected into the compressor discharge casing. The injected exhaust gas may help to increase efficiency and power output of the gas turbine system. In systems having a gas capture system (e.g., a carbon capture system), the increased efficiency and power output achieved by the injected exhaust gas helps to offset efficiency losses and power losses associated with the gas capture system. Various aspects of the EGR system and exhaust injection are discussed in further detail below.

(8) FIG. 1 is a block diagram of an embodiment of a system 10 having a gas turbine system 12 (e.g., gas turbine engine), a steam turbine system 14, a heat recovery steam generator (HRSG) 16, a gas treatment system 18 having one or more gas capture systems 20, and a controller 22 coupled to each of the systems 12, 14, 16, and 18. As discussed below, the one or more gas capture systems 20 of the gas treatment system 18 are configured to capture an undesirable gas (e.g., CO₂) from exhaust gas and/or air (e.g., by direct or indirect air capture). In the illustrated embodiment, the system 10 is a combined cycle system having the gas turbine system 12, the steam turbine system 14, and the HRSG 16. However, in some embodiments, the system 10 may be a simple cycle system having the gas turbine system 12 without the steam turbine system 14 and the HRSG 16. In either case, the system 10 includes the gas treatment system 18 having the one or more gas capture systems 20 and an exhaust gas recirculation (EGR) system 60, wherein the EGR system 60 has exhaust injection at various exhaust injection locations 59. The exhaust injection helps to increase efficiency and power output of the gas turbine system 12, thereby helping to offset efficiency and power losses associated with the one or more gas capture systems 20.

(9) Before discussing details of the gas treatment system 18, various aspects of the system 10 are discussed in further detail. For purposes of orientation in the drawings, reference may be made to an axial direction or axis 40, a radial direction or axis 42 extending radially away from the axial direction or axis 40, and a circumferential direction or axis 44 extending circumferentially around the axial direction or axis 40. The directions or axes 40, 42, and 44 may be in reference to a rotational axis of the gas turbine system 12, for example.

(10) The gas turbine system 12 includes an air intake 50, a compressor 52 having one or more compressor stages, one or more combustors 54, a compressor discharge casing 55, a turbine 56 (e.g., an expansion turbine) having one or more turbine stages, and a load 58 (e.g., electrical generator) driven by the turbine 56. As discussed above, the gas turbine system 12 includes the EGR system 60 configured to recirculate an exhaust gas 61 into the air intake 50. In certain embodiments, the EGR system 60 includes one or more cooling and cleaning module(s) 63 configured to receive exhaust gas 61 from the gas turbine system 12 and/or the HRSG 16. The cooling and cleaning modules 60 may include one or more condensers and/or one or more filters configured to cool and clean the exhaust gas 61 received from the gas turbine system 12 and/or the HRSG 16. The recirculated exhaust gas 61 helps to reduce the temperature of exhaust gas 61 and the formation of certain emissions (e.g., nitrogen oxides (NO_x)) associated with combustion

in the combustors **54**. In operation, the compressor **52** receives air (and also exhaust gas **61** if the EGR system **60** is active) from the air intake **50** and compresses the air and/or exhaust gas **61** in one or more compressor stages (e.g., stages of rotating compressor blades). The compressed air travels from the last compressor stage, through a compressor diffuser, and into a compressor discharge casing **55**, which surrounds at least the downstream portions of the combustors **54**. The combustors **54** include one or more fuel nozzles (or burners) that generate hot combustion gases by combusting fuel from a fuel supply system with the compressed air and/or exhaust gas (i.e., fluids from the compressor discharge casing **55**, which may have been used previously to cool the combustors **54**). The hot combustion gases expand and drive one or more turbine stages (e.g., stages of rotating turbine blades) in the turbine **56**, thereby driving rotation of the compressor **52** and the load **58** via shafts. The turbine **56** then outputs the energy-depleted combustion gases as the exhaust gas **61**.

(11) In a combined cycle configuration of the system **10**, the HRSG **16** recovers waste heat from the exhaust gas **61** to generate steam for driving the steam turbine system **14**. The HRSG **16** includes a high-pressure (HP) steam section **70**, an intermediate-steam (IP) section **72**, and a low-pressure (LP) steam section **74** configured to generate HP steam **76**, IP steam **78**, and LP steam **80**. The steam turbine system **14** may include an HP steam turbine **82** driven by the HP steam **76**, an IP steam turbine **84** driven by the IP steam **78**, and a LP steam turbine **86** driven by the LP steam **80**. In addition to the steam provided by the HRSG **16**, the HP steam turbine **82** provides IP steam to the IP steam turbine **84**, and the IP steam turbine **84** provides LP steam to the LP steam turbine **86**. The LP steam turbine **86** then outputs any remaining steam/water to a condensate line **88** coupled to the LP steam section **74** of the HRSG **16**. The condensate line **88** may include a condenser **90** configured to condense any remaining steam to form a condensate, and a pump **92** configured to pump the condensate back to the LP steam section **74**. In operation, the steam turbine system **14** drives a load **94** (e.g., electrical generator) via a shaft.

(12) In certain embodiments, the steam turbine system **14** and/or the HRSG **16** may provide heated water and/or steam (e.g., HP steam **76**, IP steam **78**, and/or LP steam **80**) to the gas treatment system **18** to support a desorption mode of the one or more gas capture systems **20**. For example, the gas capture systems **20** may receive heated water and/or steam in a temperature range of 100 to 150 degrees Celsius, 110 to 150 degrees Celsius, 120 to 150 degrees Celsius, or 130 to 150 degrees Celsius. In a simple cycle configuration of the system **10**, the steam turbine system **14** and the HRSG **16** are excluded from the system **10**.

(13) The exhaust gas **61** flows from the gas turbine system **12** and/or the HRSG **16** to the EGR system **60** having the cooling and cleaning module(s) **63**. In the illustrated embodiment, the EGR system **60** is configured to control flows of the exhaust gas **61** via an exhaust gas circuit **118** (e.g., fluid conduits and manifolds) having valves **128**, wherein the exhaust gas circuit **118** includes an exhaust gas circuit **130** extending to the intake **50**, an exhaust gas circuit **132** extending through an exhaust gas injection system **97** to one or more exhaust injection locations **59** downstream of the compressor **52** (e.g., compressor discharge casing **55**, combustors **54**, and/or turbine **56**), and an exhaust gas circuit **134** extending to the gas treatment system **18**. The exhaust gas circuit **130** includes a valve **128**, **131** configured to selectively control an amount or percentage of the exhaust gas **61** (e.g., intake portion **95**) directed to the intake **50**. The exhaust gas circuit **132** includes a valve **128**, **133** configured to selectively control an amount or percentage of the exhaust gas **61** (e.g., injection portion **99**) directed to the one or more exhaust injection locations **59** through the exhaust gas injection system **97**. The exhaust gas circuit **134** includes a valve **128**, **135** configured to selectively control an amount or percentage of the exhaust gas **61** (e.g., gas capture portion **107**) directed to the gas treatment system **18**. As shown, the exhaust gas circuit **130** routes the intake portion **95** of the exhaust gas **61** to the intake **50** of the gas turbine system **12**. In certain embodiments, a fan or blower may help to force a flow of the intake portion **95** of the exhaust gas **61** to the intake **50**.

(14) In the illustrated embodiment, the exhaust gas circuit **132** routes the injection portion **99** (e.g., partial extraction) of the exhaust gas **61** to the exhaust gas injection system **97**. As shown, the exhaust gas injection system **97** includes an exhaust gas compressor, such as a multi-stage compressor **93** having a plurality of stages **98**. The plurality of stages **98** is configured to compress the injection portion **99** of the exhaust gas **61** in compression stages. The multi-stage compressor **93** is configured to receive the injection portion **99** of the exhaust gas **61** from the EGR system **60**. The exhaust gas injection system **97** also includes an intercooler **101** fluidly coupled to each stage of the plurality of stages **98** of the multi-stage compressor **93**. The intercooler **101** is configured to cool the injection portion **99** of the exhaust gas **61** between the stages **98** of compression by the multi-stage compressor **93**. As shown, the exhaust gas injection system **97** also includes an engine **103** (e.g., reciprocating piston-cylinder engine) configured to drive the multi-stage compressor **93**. The engine **103** may include 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 piston-cylinder assemblies driven by combustion of a fuel, such as natural gas, diesel, gasoline, or biofuel.

(15) Additionally, the exhaust gas injection system **97** includes one or more valves **105** configured to adjust a proportion (e.g., amount) of the injection portion **99** of the exhaust gas **61** through the exhaust gas circuit **132** to the various exhaust injection locations **59** (e.g., compressor discharge casing **55**, combustors **54**, and turbine **56** of the gas turbine system **12**). Each of the exhaust injection locations **59** is generally downstream of compression (e.g., compression stages of compressor blades) of the compressor **52**. In operation, the injection portion **99** of the exhaust gas **61** is configured to increase or augment power of the gas turbine system **12**, such as by providing additional fluid energy (i.e., additional mass flow) to drive the turbine **56**. Exhaust gas **61** provided to the compressor discharge casing **55** may be used to cool the combustors **54** and may be subsequently directed to the fuel nozzles for participation in the combustion of fuel to generate additional combustion gases. Alternately, or additionally, exhaust gas **61** provided to the combustors **54** may be supplied directly to the fuel nozzles for introduction into the combustors **54** (e.g., used as a diluent or as a barrier fluid between fuel and air streams).

(16) In certain embodiments, the exhaust gas injection system **97** includes heat exchangers **109** to selectively transfer heat with (e.g., heat or cool) the injection portion **99** of the exhaust gas **61** through the exhaust gas circuit **132** to the various exhaust injection locations **59**, such that the temperature of the injection portion **99** can be controlled to generally match temperatures at the exhaust injection locations **59**. For example, the heat exchangers **109** may be configured to transfer heat between the engine **103** and the injection portion **99**, such as by circulating a heat transfer medium (e.g., lubricant, coolant, water, etc.) between the engine **103** and the heat exchangers **109**.

(17) In the illustrated embodiment, the exhaust gas circuit **134** and the valve **135** selectively route the gas capture portion **107** of the exhaust gas **61** through one or more gas capture systems **20** configured to capture undesirable gases. The undesirable gases may include carbon oxides (CO.sub.X) (e.g., carbon dioxide (CO.sub.2) and carbon monoxide (CO)), nitrogen oxides (NO.sub.X) (e.g., nitrogen dioxide (NO.sub.2)), sulfur oxides (SO.sub.X) (e.g., sulfur dioxide (SO.sub.2)), or any combination thereof. In the following discussion, CO.sub.2 may be used as an example of the undesirable gases; however, the gas capture systems **20** may be designed to capture any of the foregoing undesirable gases. For example, the gas capture systems **20** include one or more carbon capture systems **100** (e.g., CO.sub.2 capture systems). The gas capture systems **20** (e.g., carbon capture systems **100**) may include sorbent-based gas capture systems, solvent-based gas capture systems, cryogenic gas capture systems, or any combination thereof, configured to remove and capture undesirable gases. The carbon capture system **100** may include components **102**, **104**, **106**, and **108** configured to enable gas capture of undesirable gases (e.g., CO.sub.2) from the exhaust gas **61**, thereby outputting a treated gas **110** and a captured gas **112** (e.g., CO.sub.2). The treated gas **110** may be substantially free of the undesirable gases (e.g., CO.sub.2) and may be discharged through an exhaust stack. The captured gas **112** (e.g., CO.sub.2) may be compressed by a compression system **114** and stored and/or transported by a storage and/or pipeline system **116**.

(18) In certain embodiments, the carbon capture system **100** is a sorbent-based carbon capture system, and the components **102**, **104**, **106**, and/or **108** include multiple sorbent-based carbon capture units (e.g., adsorbers). For example, the sorbent-based carbon capture units may include temperature swing adsorption (TSA) units or adsorbers, wherein a temperature swing or change is used to sequentially operate in an adsorption mode, a desorption mode, and a cooling mode at different temperatures. In the adsorption mode, the adsorber is configured to adsorb undesirable gases (e.g., CO.sub.2) into sorbent material at a first temperature. In the desorption mode, the adsorber is configured to desorb the undesirable gases (e.g., CO.sub.2) from the sorbent material, for example, by heating the sorbent material from the first temperature to a higher second temperature using a heat source. The heat source may include a heated fluid, such as a heated gas and/or liquid (e.g., steam). In the cooling mode, the adsorber is cooled in preparation for the next adsorption mode.

(19) In certain embodiments, the carbon capture system **100** is a solvent-based carbon capture system, and the components **102**, **104**, **106**, and/or **108** include one or more absorbers, strippers, and associated equipment. For example, the absorber is configured to absorb undesirable gases (e.g., CO.sub.2) into a solvent, thereby outputting the treated gas **110** through an exhaust stack and a CO.sub.2-rich solvent to the stripper. The stripper is configured to apply heat to the CO.sub.2-rich solvent, thereby stripping the undesirable gases (e.g., CO.sub.2) from the solvent to produce the captured gas **112** and a CO.sub.2-lean solvent. The stripper may receive heat via a heat source, such as a heated gas and/or liquid (e.g., steam). The stripper returns the CO.sub.2-lean solvent to the absorber to repeat the cycle.

(20) In the illustrated embodiment, the controller **22** is configured to control all aspects of the system **10**. The controller **22** includes one or more processors **120**, memory **122**, instructions **124** stored on the memory **122** and executable by the processor **120**, and communication circuitry **126** configured to communicate with sensors and various equipment of the system **10**. For example, the controller **22** is configured to receive sensor feedback from sensors coupled to the gas turbine system **12**, the steam turbine system **14**, the HRSG **16**, and the gas treatment system **18** (e.g., gas capture systems **20**), and control the same equipment based on the sensor feedback, operating modes, user input, computer models, or any combination thereof. The sensors (e.g., sensors **180**, **182** in FIG. 2) may include temperature sensors, pressure sensors, flow rate sensors, gas composition sensors, or any combination thereof. In certain embodiments, the controller **22** is configured to control operation of the gas capture systems **20** (e.g., carbon capture systems **100**), such by controlling modes of operation (e.g., adsorption mode, desorption mode, and cooling mode), controlling heat sources for supplying heated fluid (e.g., steam) to the gas capture systems **20**, controlling cooling sources for supply cooled fluids to the gas capture systems **20**, or any combination thereof.

(21) In certain embodiments, the controller **22** may be configured to control flows through each of the exhaust gas circuits **118** (e.g., **130**, **132**, and **134**), such as by controlling the valves **128** (e.g., **131**, **133**, and **135**) and the valves **105**, thereby selectively varying (e.g., increasing or decreasing) exhaust flows to the intake **50**, the exhaust injection locations **59** (e.g., the compressor discharge casing **55**, the combustors **54**, and the turbine **56**), and the gas treatment system **18**. Collectively, the valves **128** (e.g., **131**, **133**, and **135**) and the valves **105** may be identified as valves **136**, wherein the valves **136** may include passive valves (e.g., check valves responsive to pressure differentials), active valves (e.g., actuator-controlled valves responsive to control by the controller **22**), or any combination thereof. Additionally, the controller **22** may be configured to control one or more components of the exhaust gas injection system **97**, such as the multi-stage compressor **93**, the intercooler **101**, the engine **103**, valves **105**, and the heat exchangers **109**.

(22) In certain embodiments, the controller **22** is configured to control the EGR system **60**, the exhaust gas injection system **97**, the gas turbine system **12**, and the gas treatment system **18** (e.g., the carbon capture system **100**) in cooperation with one another based on sensor feedback

indicative of an operating condition and/or a load on the gas turbine system **12**, power demand by the gas treatment system **18** (e.g., the carbon capture system **100**), sensed exhaust emissions versus threshold values, or any combination thereof. For example, the load on the gas turbine system **12** may include a part load condition or a full load condition, wherein the load may include the load to drive an electrical generator or other equipment, power demand on a power grid, power demand by the gas treatment system **18** (e.g., the carbon capture system **100**), or any combination thereof. The operating condition of the gas turbine system **12** may include a transient condition (e.g., a startup condition, a shutdown condition, or a variable demand condition), a steady state condition, or any combination thereof. By further example, the power demand by the gas treatment system **18** (e.g., the carbon capture system **100**) may vary depending on a concentration of undesirable gases (e.g., CO.sub.2) in the exhaust gas, a flow rate of the exhaust gases, a fuel composition used in the gas turbine system **12** (e.g., dirty versus clean fuel), a target level (e.g., percentage) of carbon capture, a number of stages of carbon capture, or any combination thereof. Similarly, the sensed emissions may include sensed emissions levels of the undesirable gases, such as CO.sub.X (CO and/or CO.sub.2), NO.sub.X, SO.sub.X, or any combination thereof, wherein the sensed emissions are compared against threshold values based on regulatory requirements, gas turbine controls, and controls by the gas treatment system **18**.

(23) In operation, the controller **22** may vary the flows of exhaust gas **61** through the intake **50** for emissions control (e.g., NO.sub.X control), through the exhaust gas injection system **97** to the one or more exhaust injection locations **59** for power augmentation, and through the exhaust gas treatment system **18** (e.g., carbon capture system **100**) for emissions control and carbon capture, such that targets are met for each of the systems. In certain embodiments, the controller **22** is configured to increase the flow of exhaust gas **61** through the exhaust gas injection system **97** to the one or more exhaust injection locations **59** to increase power augmentation to offset increases in power demand by the exhaust gas treatment system **18** (e.g., carbon capture system **100**), and the controller **22** is configured to decrease the flow of exhaust gas **61** through the exhaust gas injection system **97** to the one or more exhaust injection locations **59** to decrease power augmentation to in response to decreases in power demand by the exhaust gas treatment system **18** (e.g., carbon capture system **100**). Additionally, the controller **22** may be configured to increase or decrease the flow of exhaust gas **61** through the exhaust gas treatment system **18** (e.g., carbon capture system **100**) to increase or decrease the amount of carbon capture. Thus, the controller **22** is configured to cooperatively control the exhaust emissions levels, amounts of carbon capture, and power output to the electrical grid.

(24) FIG. 2 is a schematic of an embodiment of the system **10**, further illustrating features of the exhaust gas injection system **97** of the EGR system **60**. In the illustrated embodiment, the EGR system **60** recirculates at least a portion of the exhaust gas **61** produced by the gas turbine system **12**, wherein the EGR system **60** may receive the exhaust gas **61** directly from the gas turbine system **12** and/or indirectly via the HRSG **16**. The EGR system **60** cools and cleans the exhaust gas **61** via the cooling and cleaning module(s) **63**. The EGR system **60**, such as by controlling the valves **131**, **133**, and **135** via the controller **22**, selectively routes the exhaust gas **61** through the exhaust gas circuits **118** (e.g., **130**, **132**, and **134**) to the intake **50**, the exhaust gas injection system **97**, or the gas treatment system **18** having the carbon capture system **100**. In certain embodiments, a majority of the exhaust gas **61** is diverted to the gas treatment system **18**. For example, more than 60, 65, 70, 75, 80, or 85 percent of the exhaust gas **61** (e.g., gas capture portion **107**) may be diverted to the gas treatment system **18** along the exhaust gas circuit **134**.

(25) As shown, a remaining portion **140** of the exhaust gas **61** is diverted either to the intake **50** of the gas turbine system **12** (e.g., intake portion **95**) via the exhaust gas circuit **130** or to the exhaust gas injection system **97** (e.g., injection portion **99**) via the exhaust gas circuit **132**. In certain embodiments, the injection portion **99** of the remaining portion **140** of the exhaust gas **61** may comprise less than 15, 12, 9, 6, or 3 percent of the remaining portion **140** of the exhaust gas **61**.

That is, the majority of the remaining portion **140** is fed back to the intake **50** of the gas turbine system **12**. However, any suitable percentage or ratio of the exhaust gas **61** may be routed through the exhaust gas circuits **118** (e.g., **130**, **132**, and **134**) to the intake **50**, the exhaust gas injection system **97**, or the gas treatment system **18**. Additionally, any suitable percentage or ratio of the exhaust gas **61** may be routed through the exhaust gas circuit **118**, **132** having the exhaust gas injection system **97** to one or more of the exhaust injection locations **59** downstream of the compressor **52** (e.g., compressor discharge casing **55**, combustors **54**, and/or turbine **56**). The foregoing percentages or ratios of the exhaust gas may be achieved with the valves **128** (e.g., **131**, **133**, and **135**) and the valves **105**.

(26) As shown, the exhaust gas injection system **97** includes the multi-stage compressor **93** having the plurality of stages **98** (e.g., stages **142**, **144**, and **146**). Although the illustrated embodiment shows the multi-stage compressor **93** having three stages **98**, it should be recognized that the multi-stage compressor **93** may include two or more stages **98**. For example, the multi-stage compressor **93** may include 2, 4, 5, 6, 7, 8, or more stages **98**. The multi-stage compressor **93** is driven by the engine **103** (e.g., reciprocating piston-cylinder engine) that receives air **148** and fuel **150** (e.g., natural gas, diesel fuel, biofuel, gasoline, etc.) and produces an engine exhaust gas **152**. It may be recognized that the multi-stage compressor **93** is configured to compress the exhaust gas **61** to produce a compressed exhaust gas **156** having a pressure that substantially matches the pressure of the exhaust injection location **59**, such as the compressor discharge casing **55** of the gas turbine system **12**.

(27) The exhaust gas injection system **97** also includes the intercooler **101**. In the illustrated embodiment, the intercooler **101** is fluidly coupled to the plurality of stages **98** of the multi-stage compressor **93**. As shown, the intercooler **101** is bidirectionally fluidly coupled to the stages **142** and **144** and unidirectionally fluidly coupled to the stage **146**, such that the exhaust gas **61** flows from the intercooler **101** to the stage **146**. In certain embodiments, both the first stage (e.g., stage **142**) and the last stage (e.g., stage **146**) of the multi-stage compressor **93** may be unidirectionally fluidly coupled to the intercooler **101**. In certain embodiments, the intercooler **101** may be fluidly coupled to any combination of the stages **98** of the multi-stage compressor **93**. As shown, the intercooler **101** may receive a coolant **153** (e.g., water) from an external coolant source.

(28) In the illustrated embodiment, the exhaust gas injection system **97** also includes at least one heat exchanger **109**, such as a heat exchanger **154** (e.g., waste heat recovery system, heat exchanger, or recuperator). The heat exchanger **154** is fluidly coupled to the multi-stage compressor **93** as well as the engine **103** and is configured to receive compressed exhaust gas **156** output by the multi-stage compressor **93** and engine exhaust gas **152** output by the engine **103**. The heat exchanger **154** is configured to transfer heat from the engine exhaust gas **152** to the compressed exhaust gas **156**. After passing through the heat exchanger **154**, the compressed exhaust gas **156** may be injected into the compressor discharge casing **55** of the gas turbine system **12**. In certain embodiments, the heat exchanger **154** may be configured to heat the compressed exhaust gas **156** until a temperature of the compressed exhaust gas **156** substantially matches (e.g., within plus or minus 5, 10, 15, 20, or 25 degrees Celsius of) a temperature of the high-pressure air in the compressor discharge casing **55**. In certain embodiments, after the engine exhaust gas **152** passes through the heat exchanger **154**, the engine exhaust gas **152** may be vented into the atmosphere.

(29) In certain embodiments, the exhaust gas injection system **97** includes a turbocharger **158**. In certain embodiments, the engine **103** is fluidly coupled to the turbocharger **158**, such that the engine exhaust gas **152** is directed from the engine **103** to the turbocharger **158**, thereby at least partially powering the turbocharger **158**. The turbocharger **158** may include a turbine that is driven by the engine exhaust gas **152**. The turbocharger **158** may also include a compressor driven by the turbine that compresses the air **148** received by the engine **103**. In the illustrated embodiment, the engine **103** is fluidly coupled to the HRSG **16**, such that the engine exhaust gas **152** is received by

the HRSG **16** from the engine **103**. In certain embodiments, the engine **103** may be fluidly coupled to the compressor discharge casing **55**, such that the engine exhaust gas **152** is injected into the compressor discharge casing **55**. In certain embodiments, the engine exhaust gas **152** may be directed to the HRSG **16**, the compressor discharge casing **55**, the turbocharger **158**, or a combination thereof.

(30) In the illustrated embodiment, the exhaust gas injection system **97** is configured to inject the compressed exhaust gas **156** (e.g., the injection portion **99**) from the EGR system **60** into one or more exhaust injection locations **59** of the gas turbine system **12**. As shown, the heat exchanger **154** may be fluidly coupled to the compressor discharge casing **55** and/or a last stage portion **160** of the compressor **52** of the gas turbine system **12** (e.g., downstream from compression by compressor blades of the compressor **52**). In certain embodiments, the compressed exhaust gas **156** may be directed from the heat exchanger **154** to the last stage portion **160** of the compressor **52**, the compressor discharge casing **55**, or a combination.

(31) Additionally, or alternatively, the exhaust gas injection system **97** may be configured to inject a turbine portion **161** of the exhaust gas **61** into the turbine **56** (e.g., expansion turbine) of the gas turbine system. The exhaust gas injection system **97** may be fluidly coupled to one or more exhaust injection locations **59** (e.g., turbine locations **162**) through an exterior casing **164** of the turbine **56** into an exhaust flow path through the turbine **56**. As shown, the last stage **146** of the multi-stage compressor **93** is fluidly coupled to a first axial location **165** of the one or more turbine locations **162**, and the intermediate stage **144** of the multi-stage compressor **93** is fluidly coupled to a second axial location **166** of the one or more turbine locations **162**. The first axial location **165** and the second axial location **166** are different from one another. That is, the first axial location **165** and the second axial location **166** have different positions along the axial direction **40**. As illustrated, the first axial location **165** is upstream (i.e., forward) of the second axial location **166**. The turbine portion **161** of the exhaust gas **61** may include an upstream portion **168** that is injected from the stage **146** of the multi-stage compressor **93** into the first axial location **165** of the turbine **56**. Additionally, the turbine portion **161** of the exhaust gas **61** may include a downstream portion **170** that is injected from the stage **144** of the multi-stage compressor **93** into the second axial location **166** of the turbine **56**. It may be appreciated that the stages **98** of the multi-stage compressor **93** compress the exhaust gas **61** in discrete stages (e.g., steps), thereby enabling a substantial matching of the pressure of the exhaust gas **61** at each stage **68** with a pressure of air flowing through the gas turbine system **12**.

(32) Although the illustrated embodiment shows two stages **98** of the multi-stage compressor **93** being fluidly coupled to two turbine locations **162** of the turbine **56** of the gas turbine system **12**, it should be recognized that fewer or more stages **98** of the multi-stage compressor **93** may be fluidly coupled to the turbine **56**. For example, one stage of the multi-stage compressor **93** may be fluidly coupled to the turbine **56**. In another example, extractions from one or more stages of a multi-stage compressor **93** having 3, 4, 5, 6, 7, 8, or more stages **98** may be fluidly coupled to one or more stages of the turbine **56**. Additionally, or alternatively, the coupling of the stages **98** of the multi-stage compressor **93** with the turbine **56** may be in combination with the coupling of the heat exchanger **154** with the compressor **52** and/or the compressor discharge casing **55**.

(33) In the illustrated embodiment, the EGR system **60** and/or the exhaust gas injection system **97** includes the one or more valves **136**, which include the valves **128** (e.g., **131**, **133**, and **135**) and the valves **105** (e.g., valves **172**, **174**, **176**, **178**, **179**, and **181**). The controller **22** may be communicatively coupled to any combination of the one or more valves **136**. As shown, the valve **172** is disposed between the heat exchanger **154** and the engine **103**. The valve **172** may be configured to adjust a flow rate of the engine exhaust gas **152** from the engine **103** to the heat exchanger **154**, thereby adjusting an amount of heat transfer from the engine exhaust gas **152** to the compressed exhaust gas **156** in the heat exchanger **154**.

(34) As shown, the one or more valves **105** also include the valve **174** disposed between the heat

exchanger **154** and the intercooler **101**, such that the heat exchanger **154** is fluidly coupled to the intercooler **101**. Additionally, the valve **174** is fluidly coupled with the engine **103** and may be configured to adjust a flow rate of the engine exhaust gas **152** expelled by the heat exchanger **154** that flows into the intercooler **101**. In certain embodiments, the valve **174** may be configured to receive a signal from the controller **22** to adjust a proportion of the engine exhaust gas **152** that enters the intercooler **101** versus being vented to the air (venting not shown in FIG. 2).

(35) Additionally, an exhaust gas circuit **175** extends between the heat exchanger **154** and the gas turbine **12**. Specifically, the one or more valves **105** described above may include the valve **176** disposed along exhaust gas circuit **175** between the heat exchanger **154** and the last stage of the compressor **52** (i.e., feeding the compressor discharge casing **55**), and the valve **178** disposed between the heat exchanger **154** and the combustors **54** (e.g., directing compressed exhaust gas **156** to the fuel nozzles) along exhaust gas circuit **177**, which may be fluidly coupled to exhaust gas circuit **175**. In certain embodiments, the valve **176** may be configured to receive a signal from the controller **22** to adjust a flow rate of the compressed exhaust gas **156** being injected into the compressor discharge casing **55** from the heat exchanger **154**. Similarly, the valve **178** may be configured to receive a signal from the controller **22** to adjust a flow rate of the compressed exhaust gas **156** being injected into the combustors **54** from the heat exchanger **154**. For example, in certain embodiments, the controller **22** may instruct the valves **176** and/or **178** to decrease the flow rate of the compressed exhaust gas **156** to the compressor discharge casing **55** and/or the combustors **54** when the turbine **56** is operating at part load.

(36) Additionally, the valve **133** is disposed between the EGR system **60** and the multi-stage compressor **93**. The valve **133** may be configured to receive a signal from the controller **22** to control the proportion (e.g., the injection portion **99**) of the remaining portion **140** of the exhaust gas **61** that is diverted to the exhaust gas injection system **97**. For example, the controller **22** may be configured to instruct the valve **133** to decrease the proportion of the injection portion **99** in response to detecting that the turbine **56** is operating in part load or has ceased operation. In certain embodiments, the controller **22** may instruct the valve **133** to stop the extraction of the injection portion **99** from the exhaust gas **61** expelled from the EGR system **60** in response to the turbine **56** operating in part load. It should be recognized that the EGR system **60** may continue to operate regardless of whether the turbine **56** is operating in part load or full load.

(37) Additionally, the one or more valves **105** may include valves **179** and **181** that separately control the flow of the injection portion **99** of the exhaust gas **61** from the multi-stage compressor **93** to the turbine **56**. That is, the valve **179** may be communicatively coupled to the controller **22** and configured to adjust the flow rate of the exhaust gas **61** from the last stage **146** of the multi-stage compressor **93** to the first axial location **165** of the turbine **56**. Additionally, or alternatively, the valve **181** may be communicatively coupled to the controller **22** and configured to adjust the flow rate of the exhaust gas **61** from the intermediate stage **144** of the multi-stage compressor **93** to the second axial location **166** of the turbine **56**. The controller **22** may be configured to control the valves **179** and **181** synchronously (e.g., together) or, in certain embodiments, separately.

(38) It may be recognized that the exhaust gas injection system **97** may include a combination of the one or more valves **105** in combination with the valves **128** (e.g., **131**, **133**, and **135**). Additionally, in certain embodiments, the one or more valves **105** may include valves disposed at other locations through the exhaust gas injection system **97**. In certain embodiments, at least one of the one or more valves **105** may include a redundancy valve and/or a check valve in case of blockage.

(39) In certain embodiments, the controller **22** may be configured to control a combination of the plurality of valves **136** (e.g., **105** and **128**) based on one or more operating conditions of the system **10**. For example, the controller **22** may be configured to receive a signal indicative of a load of the gas turbine system **12**. In certain embodiments, the controller **22** may be configured to receive the signal from a sensor **180** configured to output a signal indicative of the load of the gas turbine

system **12**. The controller **22** may then determine an estimated load of the gas turbine system **12** based on the signal. For example, the controller **22** may be configured to determine whether the gas turbine system **12** is operating at part load or full load based on the received signal. The controller **22** may be configured to compare the estimated load of the gas turbine system **12** to a threshold load. For example, the threshold load may be a percentage of the full load of the gas turbine system **12**. Based on the comparison results, the controller **22** may be configured to control a combination of the plurality of valves **136** (e.g., **105** and **128**). For example, the controller **22** may be configured to control the valve **133** to divert the injection portion **99** of the exhaust gas **61** to the multi-stage compressor **93** in response to the estimated load of the gas turbine system **12** exceeding the threshold load.

(40) In certain embodiments, the controller **22** may be configured adjust at least one of the one or more valves **105** to adjust a temperature of the injection portion **99** of the exhaust gas **61**. For example, the controller **22** may be configured to receive a signal indicative of a temperature of the injection portion **99** of the exhaust gas **61** from a sensor **182**. The controller **22** may then determine an estimated temperature of the injection portion **99** of the exhaust gas **61** based on the signal. The controller **22** may be configured to compare the estimated temperature of the injection portion **99** to a threshold temperature. Based on the comparison results, the controller **22** may be configured to control a combination of the plurality of valves **105**. For example, the controller **22** may be configured to control the valve **172** to adjust an amount of engine exhaust gas **152** that is received by the heat exchanger **154**, thereby adjusting an amount by which the compressed exhaust gas **156** is heated before being injected into the compressor discharge casing **55**. In certain embodiments, controller **22** may be configured to control valve **174** to adjust an amount of engine exhaust gas **152** that is received by the intercooler **101**, thereby adjusting an amount by which the compressed exhaust gas **156** is cooled while flowing through the multi-stage compressor **93** and the intercooler **101**. In certain embodiments, the controller **22** may be configured to control a combination of the valves **172** and **174** to adjust a temperature of the injection portion **99**. Additionally, or alternatively, the controller **22** may be configured to control a combination of the valves **133**, **176**, and **178** to adjust the amount of exhaust gas **61** that is injected into the compressor discharge casing **55** and the combustors **54**.

(41) FIG. **3** is a flowchart of an example process **200** for operating the exhaust gas injection system **97** of the EGR system **60**. The process **200** may be performed by a computing device or controller disclosed above with reference to FIGS. **1** and **2** or any other suitable computing device(s) or controller(s). Furthermore, the blocks of the process **200** may be performed in the order disclosed herein or in any other suitable order. For example, certain blocks of the process **200** may be performed concurrently. In addition, in certain embodiments, at least one of the blocks of the process **200** may be omitted.

(42) In block **202** of the process **200**, a processor (e.g., controller) receives a signal indicative of a load of a gas turbine engine of a gas turbine system **12**. For example, the processor may receive a signal from a sensor, the signal indicative of a load of the gas turbine engine. The sensor may include an encoder, a torque sensor, an oxygen sensor, a fuel flowmeter, or a combination thereof.

(43) In block **204** of the process **200**, the processor determines an estimated load of the gas turbine engine **12** based on the signal. In certain embodiments, the load of the gas turbine engine **12** may be determined by comparing the flowrate of fuel at a given speed of the gas turbine engine **12** and compare the flowrate with a maximum flowrate of fuel at the same speed and in full throttle. In some embodiments, the processor determines the estimated load based on directly measuring the load using a torque sensor, an encoder, or a combination thereof.

(44) In block **206** of the process **200**, the processor compares the estimated load of the gas turbine engine to a first threshold load. In certain embodiments, the first threshold load is less than a full load (e.g., maximum load) of the gas turbine engine **12**. For example, the threshold load may be less than 50, 60, 70, 80, or 90 percent of the full load of the gas turbine engine **12**.

(45) In block **208** of the process **200**, the processor instructs an actuator (e.g., a valve) to divert a portion of an exhaust flow output by an exhaust gas recirculation (EGR) system **60** of the gas turbine engine **12** to a multi-stage compressor **93** based on the comparison results. In certain embodiments, the processor instructs the multi-stage compressor **93** to compress the exhaust flow and/or to inject the exhaust flow into the compressor discharge casing **55** of the gas turbine engine **12**. In certain embodiments, the processor instructs the multi-stage compressor **93** to compress the exhaust flow and/or to inject the exhaust flow into the one or more exhaust injection locations **59** downstream of compression by the compressor **52**.

(46) Technical effects of the disclosed embodiments include extracting a portion of the exhaust gas from the EGR system **60** of a gas turbine system **12** and injecting the portion of the exhaust gas into one or more exhaust injection locations **59** (e.g., the compressor discharge casing **55**, the combustors **54**, and/or the turbine **56** of the gas turbine system) to increase efficiency and power output of the gas turbine system **12**. By injecting the portion of the exhaust gas of the EGR system **60** into the one or more exhaust injection locations **59**, the power output and efficiency of the gas turbine system **12** is increased sufficiently to offset losses in power and efficiency attributed to operation of the gas treatment system **18**, particularly the carbon capture system **100**. It may be recognized that using a multi-stage centrifugal compressor **93** to compress the exhaust gas prior to injection into the one or more exhaust injection locations **59** may produce compressed exhaust gas 40 percent more efficiently than the axial compressor **52** of the gas turbine system **12**. Additionally, it may be recognized that injecting the exhaust gas output by the EGR system **60** into the one or more exhaust injection locations **59** (e.g., compressor discharge casing **55**, the combustors **54**, and/or the turbine **56**) increases the power output (e.g., capacity) of the gas turbine system by up to 10 percent, regardless of ambient conditions. Again, an important advantage of the disclosed embodiments includes recovering a portion of the efficiency and power lost due to the carbon capture system **100**, such that the system **10** has a relatively low, zero, or negative carbon footprint without a substantial reduction in efficiency and power output by the system **10**.

(47) The subject matter described in detail above may be defined by one or more clauses, as set forth below.

(48) A system includes: a gas turbine system having a gas turbine engine including a compressor, a combustor, and a turbine; and an exhaust gas recirculation (EGR) system configured to recirculate a first portion of an exhaust flow from the gas turbine engine to an intake of the gas turbine engine; and an exhaust gas injection system, including a multi-stage compressor including a plurality of stages, wherein the multi-stage compressor is configured to compress a second portion of the exhaust flow; and an intercooler coupled to the multi-stage compressor; wherein the exhaust gas injection system is configured to inject the second portion of the exhaust flow into the gas turbine engine at one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine.

(49) The system of the preceding clause, including a controller having a memory, a processor, and instructions stored on the memory and executable by the processor to control the exhaust gas injection system to adjust the power augmentation based on a load of the gas turbine engine, a power demand by a carbon capture system, or a combination thereof.

(50) The system of any preceding clause, including a carbon capture system configured to capture carbon from the exhaust flow from the gas turbine engine, wherein the power augmentation is configured to compensate for losses in power and/or efficiency caused by operation of the carbon capture system.

(51) The system of any preceding clause, wherein the carbon capture system includes a sorbent-based carbon capture system, a solvent-based carbon capture system, a cryogenic carbon capture system, or a combination thereof, wherein the carbon capture system is configured to capture at least carbon dioxide (CO₂) from the exhaust flow.

(52) The system of any preceding clause, including a controller having a memory, a processor, and instructions stored on the memory and executable by the processor to control the exhaust gas injection system to adjust the power augmentation based on the operation of the carbon capture system.

(53) The system of any preceding clause, wherein the controller is configured to control the exhaust gas injection system to adjust the second portion of the exhaust flow to the one or more exhaust injection locations, wherein the one or more exhaust injection locations include a last stage of the compressor, a compressor discharge casing of the compressor, the combustor, the turbine, or a combination thereof.

(54) The system of any preceding clause, wherein the one or more exhaust injection locations include a last stage of the compressor.

(55) The system of any preceding clause, wherein the one or more exhaust injection locations include a compressor discharge casing of the compressor.

(56) The system of any preceding clause, wherein the one or more exhaust injection locations include the turbine.

(57) The system of any preceding clause, wherein the one or more exhaust injection locations include a plurality of turbine injection locations into the turbine of the gas turbine engine, wherein the plurality of stages of the multi-stage compressor are coupled to the plurality of turbine injection locations.

(58) The system of any preceding clause, wherein the exhaust gas injection system includes an engine configured to drive the multi-stage compressor, wherein the engine produces an engine exhaust gas, and a heat exchanger is configured to transfer heat from the engine exhaust gas to the second portion of the exhaust flow compressed by the multi-stage compressor.

(59) The system of any preceding clause, wherein the engine includes a reciprocating piston-cylinder engine having a turbocharger driven by at least part of the engine exhaust gas.

(60) The system of any preceding clause, wherein the gas turbine system includes a heat recovery steam generator (HRSG) configured to receive at least part of the engine exhaust gas from the engine and at least part of the exhaust flow from the gas turbine engine.

(61) The system of any preceding clause, wherein the exhaust gas injection system includes a plurality of valves. The plurality of valves includes a first valve disposed between the heat exchanger and the engine, a second valve disposed between the heat exchanger and the intercooler, a third valve disposed between the heat exchanger and a compressor discharge casing of the compressor of the gas turbine engine, a fourth valve disposed between the EGR system and the multi-stage compressor, or a combination thereof.

(62) The system of any preceding clause, including a controller having a memory, a processor, and instructions stored on the memory and executable by the processor to receive a signal indicative of a load of the gas turbine engine, determine an estimated load of the gas turbine engine based on the signal, compare the estimated load of the gas turbine engine to a threshold load, and in response to the estimated load exceeding the threshold load, control the fourth valve to divert the second portion of the exhaust flow to the multi-stage compressor.

(63) A system includes a controller having a memory, a processor, and instructions stored on the memory and executable by the processor to control a first portion of an exhaust flow from a gas turbine engine through an exhaust gas recirculation (EGR) system to an intake of the gas turbine engine, wherein the gas turbine engine includes a compressor, a combustor, and a turbine. The controller is further configured to control a second portion of the exhaust flow from the gas turbine engine through an exhaust gas injection system to one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine.

(64) The system of the preceding clause, wherein the controller is configured to adjust the power augmentation based on a load of the gas turbine engine, a power demand by a carbon capture

system, or a combination thereof.

(65) The system of any preceding clause, wherein the exhaust gas injection system includes a multi-stage compressor having a plurality of stages, and an intercooler coupled to the multi-stage compressor.

(66) A method including controlling a first portion of an exhaust flow from a gas turbine engine through an exhaust gas recirculation (EGR) system to an intake of the gas turbine engine, wherein the gas turbine engine includes a compressor, a combustor, and a turbine. The method further includes controlling a second portion of the exhaust flow from the gas turbine engine through an exhaust gas injection system to one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine.

(67) The method of the preceding clause, wherein controlling the second portion of the exhaust flow includes adjusting the power augmentation based on a load of the gas turbine engine, a power demand by a carbon capture system, or a combination thereof.

(68) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A system, comprising: a gas turbine system comprising: a gas turbine engine comprising a compressor, a combustor, and a turbine; and an exhaust gas recirculation (EGR) system configured to recirculate a first portion of an exhaust flow from the gas turbine engine to an intake of the gas turbine engine; an exhaust gas injection system, comprising: a multi-stage compressor comprising a plurality of stages, wherein the multi-stage compressor is configured to compress a second portion of the exhaust flow; an intercooler coupled to the multi-stage compressor; and a heat exchanger configured to transfer heat from a combustion exhaust gas output by a combustion system to the second portion of the exhaust flow compressed by the multi-stage compressor, wherein the combustion system is different from the combustor of the gas turbine system; wherein the exhaust gas injection system is configured to inject the second portion of the exhaust flow into the gas turbine engine at one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine.

2. The system of claim 1, comprising a controller having a memory, a processor, and instructions stored on the memory and executable by the processor to control the exhaust gas injection system to adjust the power augmentation based on a load of the gas turbine engine, a power demand by a carbon capture system, or a combination thereof.

3. The system of claim 1, comprising a carbon capture system configured to capture carbon from the exhaust flow from the gas turbine engine, wherein the power augmentation is configured to compensate for losses in power and/or efficiency caused by operation of the carbon capture system.

4. The system of claim 3, wherein the carbon capture system comprises a sorbent-based carbon capture system, a solvent-based carbon capture system, a cryogenic carbon capture system, or a combination thereof, wherein the carbon capture system is configured to capture at least carbon dioxide (CO₂) from the exhaust flow.

5. The system of claim 3, comprising a controller having a memory, a processor, and instructions

stored on the memory and executable by the processor to control the exhaust gas injection system to adjust the power augmentation based on the operation of the carbon capture system.

6. The system of claim 5, wherein the controller is configured to control the exhaust gas injection system to adjust the second portion of the exhaust flow to the one or more exhaust injection locations, wherein the one or more exhaust injection locations comprise a last stage of the compressor, a compressor discharge casing of the compressor, the combustor, the turbine, or a combination thereof.

7. The system of claim 1, wherein the one or more exhaust injection locations comprise a last stage of the compressor.

8. The system of claim 1, wherein the one or more exhaust injection locations comprise a compressor discharge casing of the compressor.

9. The system of claim 1, wherein the one or more exhaust injection locations comprise the turbine.

10. The system of claim 9, wherein the one or more exhaust injection locations comprise a plurality of turbine injection locations into the turbine of the gas turbine engine, wherein the plurality of stages of the multi-stage compressor are coupled to the plurality of turbine injection locations.

11. The system of claim 1, wherein the combustion system comprises an engine configured to drive the multi-stage compressor, wherein the engine produces the combustion exhaust gas.

12. The system of claim 11, wherein the engine comprises a reciprocating piston-cylinder engine having a turbocharger driven by at least part of the combustion exhaust gas.

13. The system of claim 11, wherein the gas turbine system comprises a heat recovery steam generator (HRSG) configured to receive at least part of the combustion exhaust gas from the engine and at least part of the exhaust flow from the gas turbine engine.

14. The system of claim 11, wherein the exhaust gas injection system comprises a plurality of valves, wherein the plurality of valves comprises: a first valve disposed between the heat exchanger and the engine; a second valve disposed between the heat exchanger and the intercooler; a third valve disposed between the heat exchanger and a compressor discharge casing of the compressor of the gas turbine engine; a fourth valve disposed between the EGR system and the multi-stage compressor; or a combination thereof.

15. The system of claim 14, comprising a controller having a memory, a processor, and instructions stored on the memory and executable by the processor to: receive a signal indicative of a load of the gas turbine engine; determine an estimated load of the gas turbine engine based on the signal; compare the estimated load of the gas turbine engine to a threshold load; and in response to the estimated load exceeding the threshold load, control the fourth valve to divert the second portion of the exhaust flow to the multi-stage compressor.

16. A system, comprising: a controller having a memory, a processor, and instructions stored on the memory and executable by the processor to control: a first portion of an exhaust flow from a gas turbine engine through an exhaust gas recirculation (EGR) system to an intake of the gas turbine engine, wherein the gas turbine engine comprises a compressor, a combustor, and a turbine; and a second portion of the exhaust flow from the gas turbine engine through an exhaust gas injection system to one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine; wherein the controller is configured to adjust the power augmentation based on an operation of a carbon capture system.

17. The system of claim 16, wherein the controller is configured to adjust the power augmentation to compensate for losses in power and/or efficiency caused by the operation of the carbon capture system.

18. The system of claim 16, comprising a heat exchanger configured to transfer heat from a combustion exhaust gas output by a combustion system to the second portion of the exhaust flow, wherein the combustion system is different from the combustor of the gas turbine engine.

19. A method, comprising: controlling a first portion of an exhaust flow from a gas turbine engine

through an exhaust gas recirculation (EGR) system to an intake of the gas turbine engine, wherein the gas turbine engine comprises a compressor, a combustor, and a turbine; controlling a second portion of the exhaust flow from the gas turbine engine through an exhaust gas injection system to one or more exhaust injection locations to provide power augmentation of the gas turbine engine, the one or more exhaust injection locations being axially spaced from the intake of the gas turbine engine; and adjusting the power augmentation based on an operation of a carbon capture system.

20. The method of claim 19, comprising transferring heat, via a heat exchanger, from a combustion exhaust gas output by a combustion system to the second portion of the exhaust flow, wherein the combustion system is different from the combustor of the gas turbine engine.
